

Ovule-number project: data structure, model-readiness, and analysis plan

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Purpose

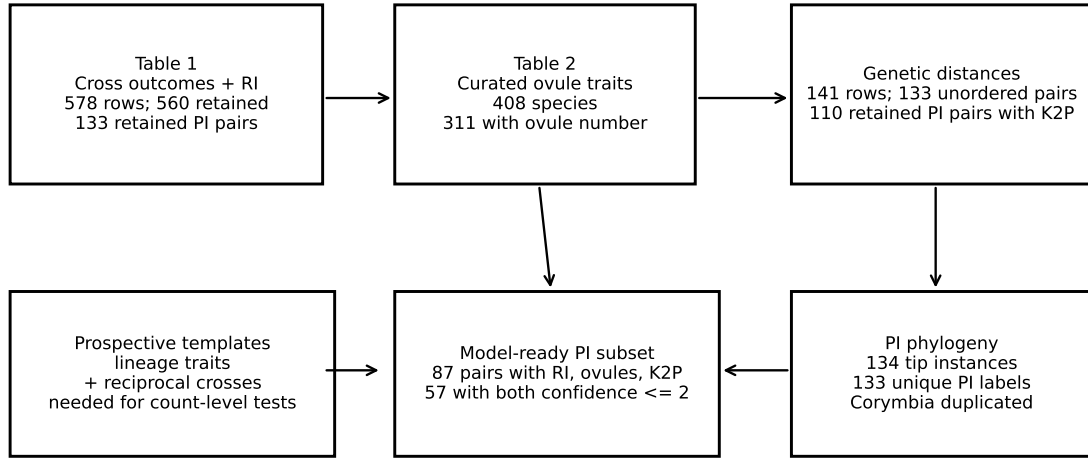
This report characterises the files supplied for the ovule-number project and asks a practical question: which parts of the data can test a model linking ovule number, reproductive isolation, genetic distance, life history, mating system, and phylogenetic structure? I treat Table 1 as the pairwise reproductive-isolation and cross-outcome table, Table 2 as the curated lineage-level covariate table, Table 2b as the master ovule-number database, the genetic-distance table as pair-level divergence information, and the Newick file as the phylogenetic structure for the phylogenetically independent pairs.

The working model assumed here is not a single fitted equation. It is the data structure needed to test whether ovule number, or contrasts in ovule number between species, predict cross success or reproductive isolation once genetic distance, mating system, life history, measurement type, and phylogenetic non-independence are controlled. This matters because the same biological idea can be tested in two different ways. A pair-level test asks whether species pairs with larger ovule-number contrasts have larger RI. A directional test asks whether the outcome differs when the many-ovule species is the maternal parent rather than the paternal parent. The second test is closer to the mechanism, but it needs reciprocal cross data on the right scale.

Main result. The data are already close to a useful comparative analysis. After removing rows flagged for removal, there are 560 retained Table 1 rows and 133 retained rows with <code>PI_name</code>. The most conservative complete-case subset with PI status, both ovule numbers, and nonmissing K2P distance contains 87 pairs. If both ovule estimates are restricted to confidence categories 1–2, this decreases to 57 pairs.
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Table 1: Input files and their modelling roles.

dataset	role	rows	columns	families	genera	unique_species
Table1.2026.01.09.csv	Pairwise cross outcomes and RI	578	15	50	93	410
Table2.2025.12.15.csv	Curated ovule, life-history, and mating-system covariates	408	6		94	408
Table2b.2025.11.25.csv	Ovule-number master table before final curation	5496	7	309	1840	3789
Genetic_distances_results.csv	K2P genetic distances for PI species pairs	141	3			266
All.PI_pairs_Zuntini.2025.10.tre	Phylogeny over PI-pair labels	134				133

Data structure and joins for the ovule-number project**Figure 1:** Overview of the supplied data files and the main joins needed for model testing.

Data inventory

Table 1 contains the response variables: reciprocal intraspecific and interspecific cross outcomes, hybrid proportions for pollen-competition rows, RI, and measurement type. Table 2 contains the main lineage-level predictors: ovule number, confidence in the estimate, life history, and mating system. Table 2b is valuable as provenance and as a source for future curation, but Table 2 is the better analysis table because it supplies one best estimate per species.

A small naming issue should be fixed before formal analysis. The column-description file refers to a PI column, but the uploaded Table 1 does not contain a separate PI column. It contains `PI_name`. I therefore treated a nonmissing `PI_name` as the operational PI flag in this audit. That is reasonable for now because the retained PI rows and the tree labels agree, but I would add an explicit Boolean column called `is_PI` before modelling.

Coverage and model readiness

The most important structural result is the cascade from all retained cross rows to complete model-ready rows. Table 2 matches all retained Table 1 species by exact binomial names, but not all Table 2 rows have a numeric ovule estimate. The genetic-distance file covers the PI pairs, but 23 rows

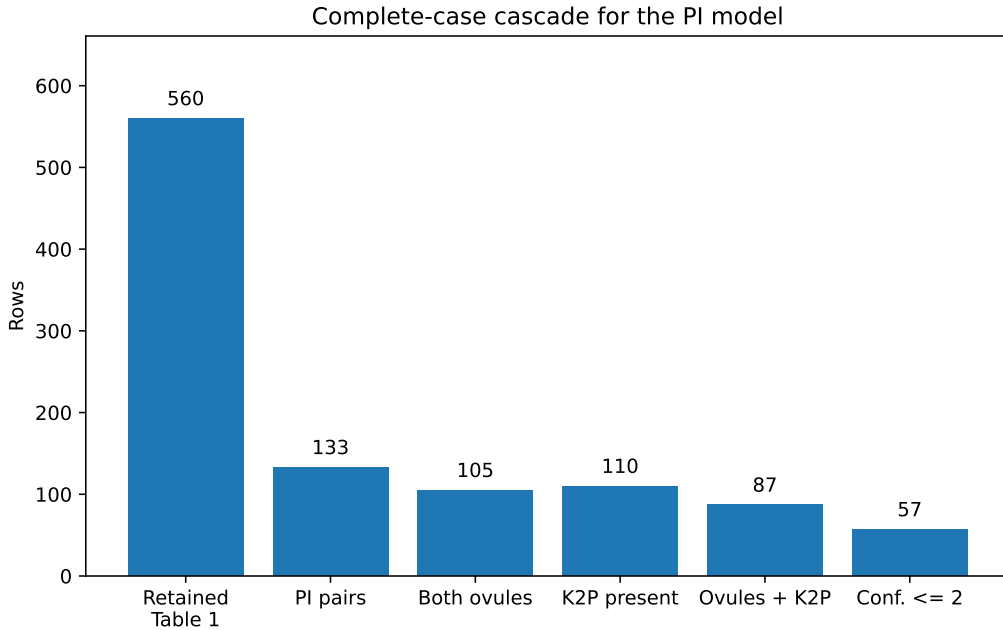
Table 2: Important structural features of Table 1.

quantity	count	comment
All pair rows	578	Before applying remove flag
Retained pair rows	560	remove == no
Rows flagged for removal	18	Duplicates or lower-quality alternatives
Rows with PI.name	141	Available proxy for PI flag
Retained PI rows	133	remove == no and PI.name present
Unique PI names retained	133	Should map to phylogeny tips
Negative RI values	59	Interspecific performance can exceed intraspecific baseline
RI values equal to 1	143	Boundary values affect beta regression

Table 3: Join coverage across cross, ovule, and genetic-distance data.

subset	rows	both_species_in_Table2	both_ovules_numeric	K2P_row_present	K2P_nonmissing	both_ovules_and_K2P	both_ovules_K2P.conf.le.3	both_ovules_K2P.conf.le.2
All Table1 rows	578	577	448	141	118	94	93	61
Retained Table1 rows	560	560	434	133	110	87	86	57
Retained PI rows	133	133	105	133	110	87	86	57

have missing K2P values. These two sources of missingness define the effective sample size for the strongest test.

**Figure 2:** The effective sample size after imposing phylogenetic independence, ovule-number availability, K2P availability, and high-confidence ovule estimates.

The retained PI subset contains 133 rows. Of these, 105 have both ovule numbers and 110 have nonmissing K2P distances. The intersection is 87 rows. This means a complete-case PI analysis is feasible, but it will have limited power for interactions among ovule number, genetic distance, mating system, life history, and measurement type. The model should therefore start simple and add complexity only when the residual structure justifies it.

Measurement type is a major source of heterogeneity. Fruit set, seed set, seed number per fruit, and pollen competition are not the same response. They differ in the biological process measured and in the denominator. Fruit set measures whether a fruit forms. Seed set includes failed crosses when expressed as seeds per pollination. Seed number per fruit excludes failed crosses. Pollen competition

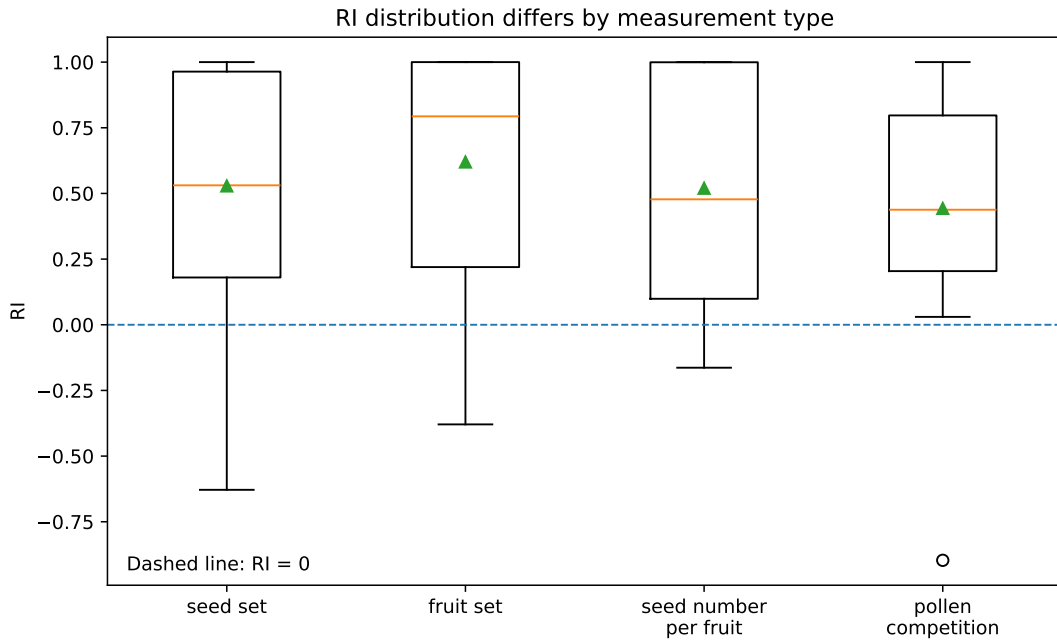
Table 4: Readiness by cross-outcome measure.

measure	retained_rows	PI_rows	both_ovules	nonmissing_K2P	complete_PI_K2P_ovules	RI_mean	RI_median	RI_min	RI_max
fruit set	146	26	119	24	20	0.62	0.794	-0.379	1
pollen competition	18	12	14	11	9	0.443	0.438	-0.897	1
seed number per fruit	77	13	54	12	9	0.52	0.478	-0.164	1
seed set	319	82	247	63	49	0.529	0.531	-0.628	1

Table 5: Ovule-number coverage by confidence category.

confidence	species	ovule_numbers_present	ovule_numbers_missing	median_ovules	max_ovules
1	208	208	0	21.56	9478
2	16	16	0	4	4
3	85	85	0	106	2000
4	67	2	65	3500000	3500000
5	32	0	32		

uses hybrid proportions rather than direct seed or fruit counts. Pooling these four responses without a measurement-type term would mix mechanisms and can create regression slopes that are hard to interpret.

**Figure 3:** RI values by measurement type among retained Table 1 rows. The distributions differ enough that measurement type should be modelled, stratified, or treated as a sensitivity factor.

Ovule-number data

Table 2 contains 408 species. Numeric ovule estimates are present for 311 species and missing for 97. Confidence is useful because it gives a simple way to separate species-level estimates from weaker genus-level or indirect estimates. Most missing ovule numbers are in confidence categories 4 and 5.

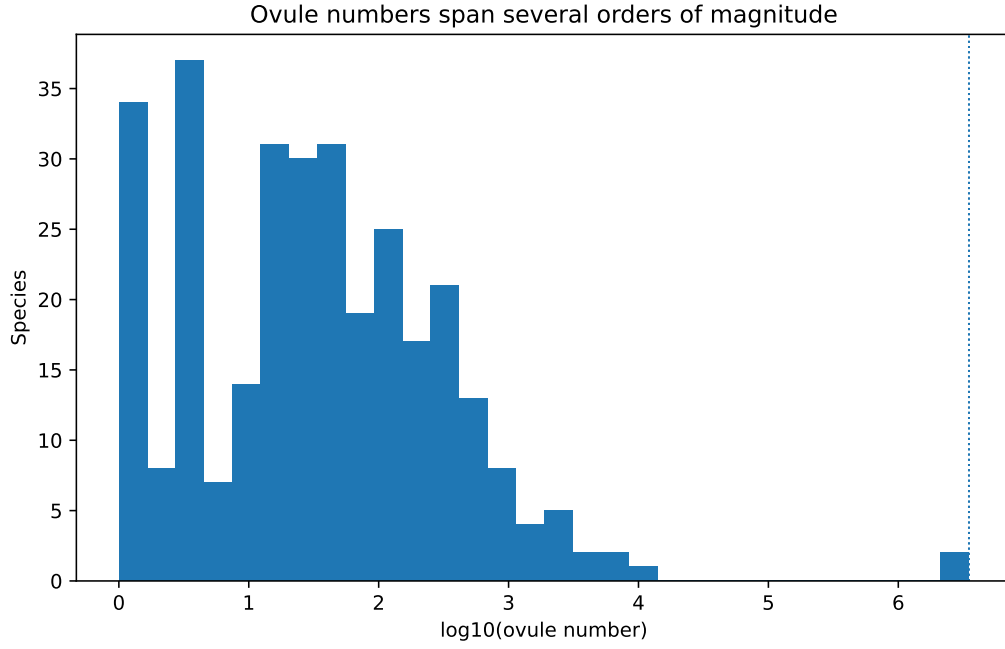


Figure 4: Ovule numbers are highly skewed, so model terms based on ovule number should use a log scale. The two largest estimates are low-confidence values and should be checked in sensitivity analyses.

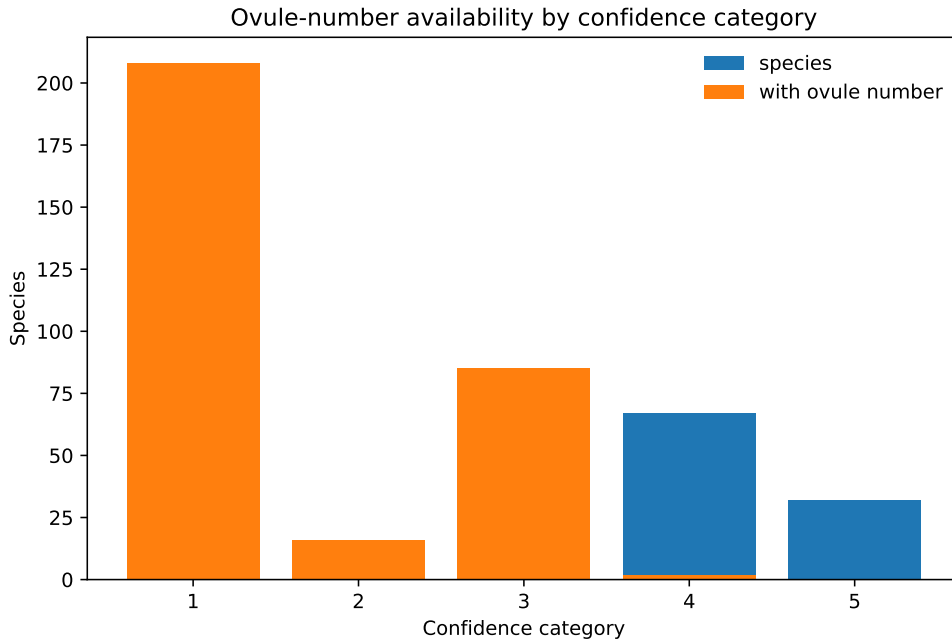


Figure 5: Ovule-number availability by confidence category. Confidence categories 1–3 carry nearly all numeric ovule estimates.

The main modelling consequence is simple. Raw ovule number should not enter the model linearly. It spans several orders of magnitude, and a few taxa can dominate raw-scale fits. Use $\log_{10}(O)$, where O is ovule number, and then construct biologically interpretable contrasts such as

$$\Delta O = |\log_{10}(O_1) - \log_{10}(O_2)|,$$

for pair-level analyses, and

$$D_{m,p} = \log_{10}(O_m) - \log_{10}(O_p),$$

for directional analyses, where m and p denote maternal and paternal species.

Table 6: Largest ovule-number estimates in Table 2.

species	ovule_number	confidence	life_history	mating_system
<i>Cattleya sincorana</i>	3500000	4	perennial	outcrossing
<i>Cattleya pfisteri</i>	3500000	4	perennial	outcrossing
<i>Orchis provincialis</i>	9478	1	perennial	mixed-mating
<i>Dactylorhiza incarnata</i>	7756	1	perennial	mixed-mating
<i>Platanthera chlorantha</i>	5831	1	perennial	mixed-mating
<i>Neotinea tridentata</i>	4452	1	perennial	outcrossing
<i>Platanthera bifolia</i>	4004	1	perennial	mixed-mating
<i>Dactylorhiza maculata</i>	2605.178	1	perennial	mixed-mating
<i>Nicotiana longiflora</i>	2367.92	1	perennial	mixed-mating
<i>Erythranthe lutea</i>	2140.32	1	perennial	mixed-mating
<i>Cyrtandra kauaiensis</i>	2000	3	woody	outcrossing
<i>Gymnadenia conopsea</i>	2000	3	perennial	mixed-mating
<i>Nicotiana alata</i>	1732	3	perennial	outcrossing
<i>Cyrtandra platyphylla</i>	1700	3	woody	outcrossing
<i>Nicotiana plumbaginifolia</i>	1538.48	1	annual	selfing

Table 7: Summary of genetic-distance data.

quantity	count	comment
Rows in genetic-distance table	141	
Unique unordered species pairs	133	Duplicates are identical for eight pairs
Rows with missing K2P	23	
Rows with nonmissing K2P	118	
Median nonmissing K2P	0.012	
Maximum nonmissing K2P	0.291	

Genetic distance and phylogeny

The genetic-distance table has 141 rows and 133 unique unordered species pairs. Eight species pairs are duplicated in the file; the duplicated distances are identical in this audit, so they do not change the values, but they should still be collapsed before modelling. The larger issue is missing distance values: 23 rows have no K2P value. Because the retained PI cross rows all have a corresponding distance-table row, the missingness is not a join problem. It is a missing-value problem.

The tree is nearly aligned with the PI structure. All retained `PI_name` labels are present in the tree. However, the tree contains 134 tip instances but only 133 unique tip labels; the duplicated tip label is *Corymbia*. Many phylogenetic comparative tools require unique tip labels. Before fitting

Table 8: Summary of PI-name matching to the phylogeny.

metric	value
tip instances	134
unique tip labels	133
duplicated tip labels	1
PI names in clean Table1	133
PI names found in tree	133
PI names missing from tree	0

phylogenetic models, this duplicate label should be resolved, either by renaming the tips with unique suffixes or by checking whether one occurrence is unintended.

What the current data can test

The pair-level test is ready for a first analysis. The clean dataset `model_ready_PI_pairs.csv` contains 87 rows with RI, both ovule numbers, and K2P distance. A useful first model would be

$$RI_i = \alpha + \beta_1 \log_{10}(d_i + c) + \beta_2 \Delta O_i + \beta_3 \bar{O}_i + \gamma_{measure(i)} + \varepsilon_i,$$

where d_i is genetic distance, c is a small constant if zero distances are retained, ΔO_i is the absolute log ovule-number contrast, $\bar{O}_i$ is the mean log ovule number of the pair, and $\gamma_{measure(i)}$ is a measurement-type effect. I would not start with all life-history and mating-system interactions because the complete-case PI subset is too small for that many terms.

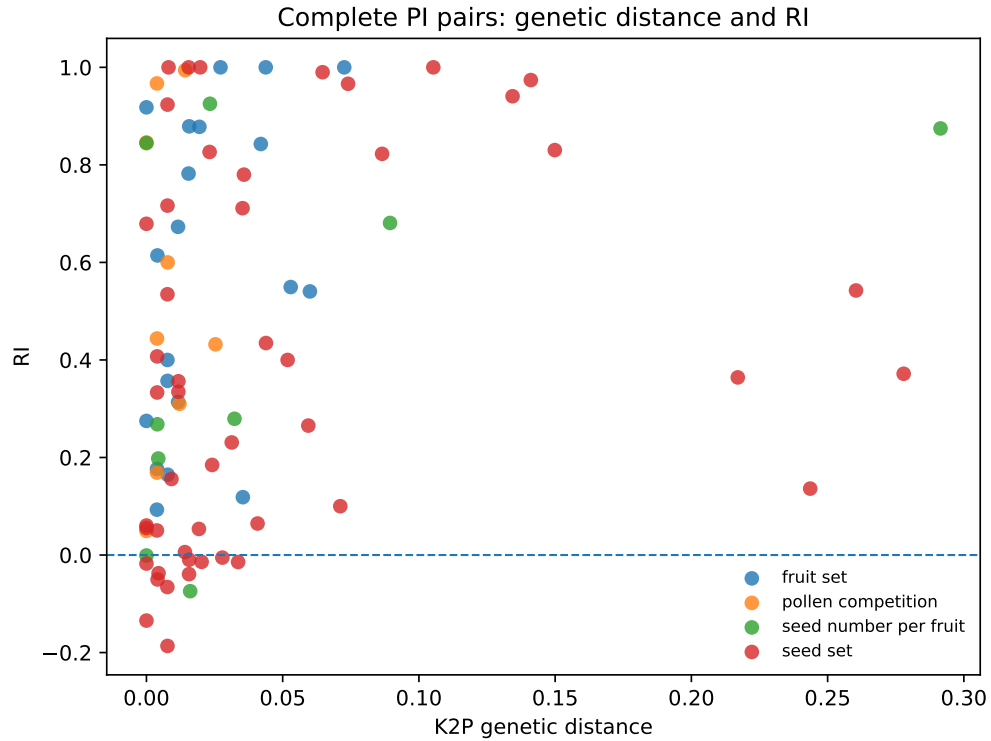


Figure 6: Exploratory relationship between K2P distance and RI in the complete PI subset. This is a diagnostic plot, not a fitted model.

Table 9: Descriptive correlations in the retained PI complete-case subset; these are diagnostics, not model tests.

x	y	n	pearson	spearman
RI	K2P_Distance	87	0.206	0.322
RI	mean_log10_ovules	87	-0.14	-0.123
RI	abs_log10_ovule_difference	87	-0.116	-0.214
K2P_Distance	abs_log10_ovule_difference	87	-0.166	-0.188

Table 10: Readiness of the reciprocal-direction data structure.

subset	directions	complete_without_K2P	complete_with_K2P	complete_PI_with_K2P	negative_directional_RI
Retained Table1 directional rows	1120	682	132	132	101

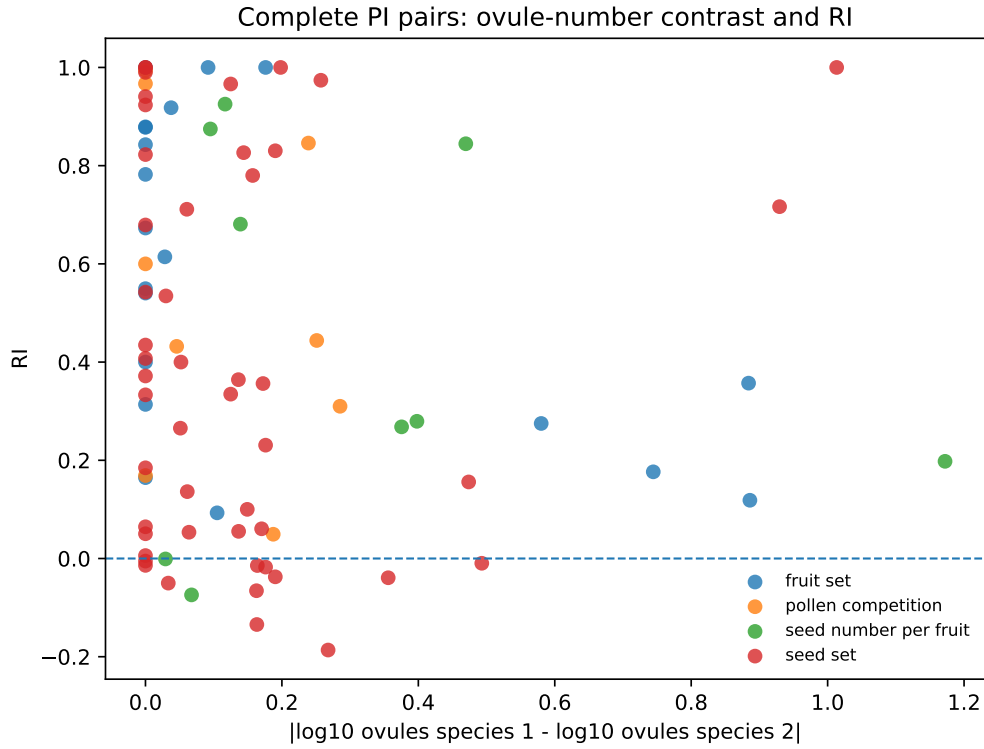


Figure 7: Exploratory relationship between absolute log ovule-number contrast and RI in the complete PI subset.

The directional test is biologically attractive because ovule number belongs to the maternal flower. For example, if the model predicts that a many-ovule maternal parent buffers cross failure, or alternatively exposes stronger post-pollination sorting, the reciprocal direction matters. The audit creates a file called `directional_cleaned_crosses.csv` with one row per maternal direction.

The directional structure is promising but uneven. In the retained data there are 1120 possible reciprocal directions, of which 682 have the interspecific outcome, maternal intraspecific baseline, and both ovule numbers. Only 132 also have K2P distance. Directional RI can be calculated as

$$RI_{m,p} = 1 - \frac{C_{m,p}}{C_{m,m}},$$

where $C_{m,p}$ is the cross outcome with maternal species m and paternal species p , and $C_{m,m}$ is the intraspecific baseline for the maternal species. This definition is useful for diagnostics, but it

inherits all problems of ratio data. It can be negative, it can exceed one if denominators are small or inconsistent, and it loses the original sampling variance.

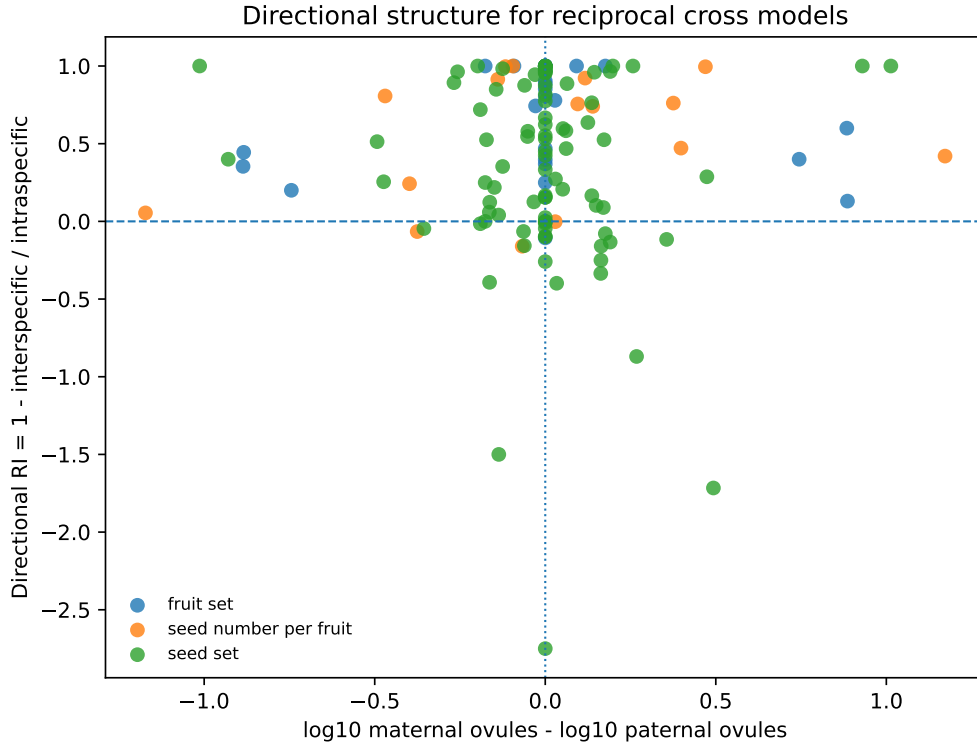


Figure 8: Directional RI against the signed maternal-paternal ovule-number contrast. The vertical line marks equal ovule numbers between maternal and paternal species.

Strengths of the datasets

First, the project already separates response data from lineage-level covariates. That is good practice. Table 1 carries the pairwise RI and cross outcomes, while Table 2 carries ovule number, confidence, life history, and mating system. This separation makes joins explicit and allows uncertainty in ovule estimates to be modelled rather than hidden.

Second, the PI structure is coherent. The retained `PI_name` values in Table 1 match the tree labels. This supports a conservative primary analysis based on the PI subset. It also makes it possible to compare three approaches: all retained rows, retained PI rows, and phylogenetic models using the full tree.

Third, reciprocal cross columns are present. This is important because ovule number is maternal in its immediate mechanism. A pair-level RI index can test whether species-pair contrasts matter, but reciprocal cross data can test whether the maternal side of the contrast matters.

Fourth, confidence categories are available for ovule number. This is useful for sensitivity analysis. A model fitted to all numeric ovule estimates can be compared with a stricter model using only confidence categories 1–2 or 1–3.

Fifth, the prospective templates are pointed in the right direction. The lineage template includes ovules, pollen production, pollen load, stigma size, style length, mating system, pollination syndrome, and flowering time. The cross template includes flower counts, ovules exposed, seeds matured, fruit set, pollen dose, pollen-tube information, F1 viability, F1 fertility, and genetic distance. Those fields would allow a much stronger count-level model than the current aggregated RI table.

Weaknesses and consequences for the model

The main weakness is that the effective sample size drops when the biologically strongest conditions are imposed. A PI complete-case model with both ovule numbers and K2P distance has 87 rows. This is sufficient for a simple analysis, but not for a saturated model with multiple interactions and random effects. A good first analysis should therefore test a small number of contrasts: genetic distance, absolute ovule-number difference, mean ovule number, and measurement type.

The second weakness is measurement heterogeneity. RI values derived from fruit set, seed set, seed number per fruit, and pollen competition do not share the same denominator. This affects the meaning of the slope. If ovule number predicts seed number per fruit but not fruit set, then pooling the responses may obscure the mechanism. Treat measurement type as a fixed effect, fit stratified models as sensitivity checks, or use a hierarchical model with measure-specific intercepts and slopes.

The third weakness is the loss of denominator information. Aggregated RI values hide the number of flowers, ovules, fruits, or seeds behind each estimate. This matters because an RI value based on 20 crosses should not carry the same precision as an RI value based on 500 crosses. The future cross-level template solves this. For the current data, use robust standard errors, nonparametric bootstrap over pairs, and sensitivity analyses by measurement type and confidence.

The fourth weakness is missing and uncertain ovule data. Table 2 has 97 missing ovule numbers, and some large estimates have low confidence. The practical consequence is that complete-case analyses may be biased if missingness is phylogenetically or biologically structured. Use confidence-weighted sensitivity analyses and consider multiple imputation, but do not let imputation create false precision. Report complete-case and imputed results side by side.

The fifth weakness is the response scale. The RI column ranges from -0.8968 to 1.0000, with 59 negative values. Negative values are biologically meaningful because they indicate interspecific performance exceeding the intraspecific baseline. Boundary values at 1 are also present. A standard beta regression is therefore not valid on raw RI without transformation or censoring. Prefer models on log relative success where possible, or use Gaussian/robust models for RI with careful residual checks.

The sixth weakness is tree-label duplication. A duplicated *Corymbia* tip may break phylogenetic covariance construction. This is easy to fix, but it must be fixed before using tools such as *ape*, *phylolm*, *MCMCglmm*, *brms*, or *RevBayes*.

Suggested analysis tools

For the current pair-level data, start with a transparent R or Python pipeline that creates a single analysis table. The minimum columns are species pair, `PI_name`, RI, measure, both ovule numbers, ovule confidence categories, life history, mating system, and K2P distance. The file `model_ready_PI_pairs.csv` created by this audit is a first version of that table.

For simple first-pass models, use linear models or robust linear models on RI, with ΔO , mean log ovule number, K2P distance, and measurement type as predictors. This gives a readable baseline. Follow this with permutation or bootstrap intervals over PI pairs. The bootstrap should resample pairs, not rows nested within the same pair.

For phylogenetic sensitivity, use phylogenetic generalised least squares. In R, *ape* can read and clean the Newick tree, *caper* or *nlme* can fit PGLS-style models, and *phylolm* can fit phylogenetic linear models. If you need measurement-error models or crossed random effects, use *brms* with a phylogenetic covariance matrix. The PI subset should remain the primary check because it is easy to explain.

Table 11: Summary of the ovule-number master table.

quantity	count
Rows in master table	5496
Rows marked table1 == yes	1208
Rows marked for removal	832
Nonremoved rows	4664
Nonremoved rows with ovule number	4206
Unique nonremoved species names	3779
Unique nonremoved table1 species names with number	711

For reciprocal cross models, reshape the data into maternal-direction rows. A useful diagnostic response is $\log(C_{m,p} + c) - \log(C_{m,m} + c)$, where c is a small offset chosen before looking at the results. The key predictor is $D_{m,p} = \log_{10}(O_m) - \log_{10}(O_p)$. Include pair as a clustering unit. If raw counts become available, replace the ratio response with a binomial, beta-binomial, Poisson, negative-binomial, or hurdle model depending on the denominator and outcome.

For missing ovule numbers, first report complete-case results. Then run sensitivity analyses using confidence thresholds: all numeric estimates, confidence 1–3, and confidence 1–2. Multiple imputation can be useful, but it should include genus, family, life history, mating system, and phylogenetic information. Imputation should not be used as the only analysis.

For future data collection, use the supplied templates but keep count denominators. The most useful fields are flowers crossed, ovules exposed, seeds matured, fruits formed, pollen dose, pollen tubes reaching the ovary, F1 viability, and F1 fertility. These allow the model to distinguish three stages: pollination success, seed maturation, and later hybrid fitness. That distinction is where the ovule-number model can become mechanistic rather than purely comparative.

Recommended next steps

The immediate next step is to verify the PI flag. Add an explicit `is_PI` column to Table 1 and keep `PI_name` as the tree label. Then collapse duplicate genetic-distance rows and decide how to treat the 23 missing K2P values. Next, inspect the largest ovule-number estimates, especially low-confidence values, because they can dominate log-scale diagnostics less than raw-scale fits but still affect leverage.

After that, fit three models in order. First, fit a complete-case PI pair-level model with RI as the response. Second, repeat the model under stricter ovule-confidence thresholds. Third, fit a directional reciprocal-cross model using `directional_cleaned_crosses.csv`. The directional model is the most biologically informative, but the pair-level model is the safest first test.

Bottom line. The datasets are strong enough for a careful first paper-level analysis, provided the claims are framed around a conservative PI complete-case test and sensitivity analyses. The strongest future version of the project would move from aggregated RI to count-level reciprocal crosses, because ovule number is a denominator and a maternal trait, not just a lineage-level covariate.

Table 12: The 20 columns with nonzero missingness shown first by dataset.

dataset	column	missing	missing_percent
Genetic distances	K2P_Distance	23	16.3
Table1	sp2_hybrid_proportion	559	96.7
Table1	sp1_hybrid_proportion	556	96.2
Table1	PI_name	437	75.6
Table1	interspecific_value_sp2_sp1	145	25.1
Table1	intraspecific_value_sp2	121	20.9
Table1	interspecific_value_sp1_sp2	87	15.1
Table1	intraspecific_value_sp1	73	12.6
Table2	ovule_number	97	23.8
Table2b	Remove	4664	84.9
Table2b	number	468	8.5
Table2b	family	11	0.2
Table2b	genus	11	0.2
Table2b	level	11	0.2

Additional structural tables

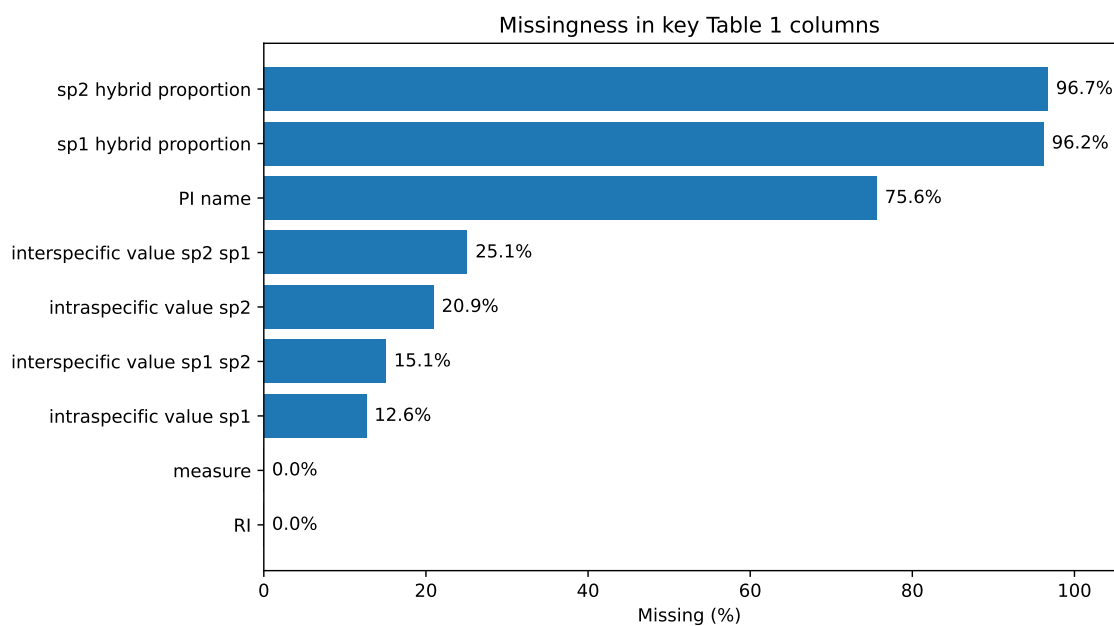


Figure 9: Missingness in key Table 1 fields. Hybrid-proportion columns are sparse because they apply mainly to pollen-competition rows.

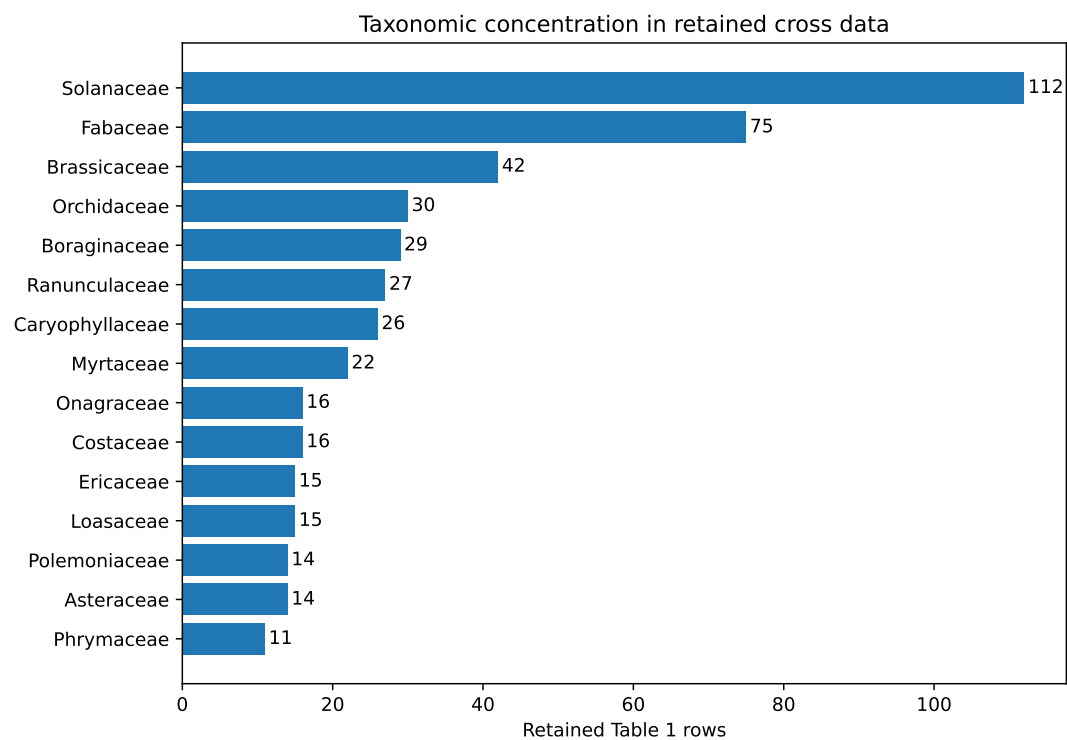


Figure 10: Retained Table 1 rows are concentrated in a subset of plant families. This should be considered when interpreting broad comparative claims.